

A Tale of Two Conferences

Hakan Erdogmus

The International Conference on Software Engineering and the Agile (formerly Agile/XP Universe) series have been two constants in my annual conference circuit since 2001. Both are excellent events in their own right, but I also enjoy them for the contrast they provide. ICSE is mainly an academic conference, the playground of applied software engineering researchers. Agile, on the other hand, is mainly a practitioner's event, the playground of developers, testers, software consultants, and, more recently, software project managers. I've always been intrigued by the differences in the two audiences' perspectives. Both perspectives are valuable, but I see the differences to be especially revealing.



For the 2008 ICSE and Agile meetings, I prepared a flash poll to gauge key perceptions regarding the separation between software engineering research and software development practice. I tweaked and transposed the questions as appropriate to suit each conference's profile. Armed

with tablet PCs, two volunteers hit the hallways of the two conferences' venues on a pulse-checking mission.

The Conference Profiles

ICSE is the largest international, and one of the two top-rated, academic conferences in software engineering. It also features an industry track. With 2008 conference attendance at more than 1,100, we were able to survey 150 delegates, for a sampling rate of 13 percent.

Figure 1 shows our poll results. Of the respondents, 89 percent identified their role as researcher. Only 33 percent identified themselves as practitioners. These groups included the 23 percent of the respondents who identified themselves as both practitioners and researchers. Sixty-six percent had an academic affiliation, 23 percent had an industry affiliation, and 11 percent had a government affiliation.

Agile 2008 is predominantly a practitioner's conference, but it does feature research tracks, including technical papers and an emerging-results session. In 2008, attendance exceeded 1,700 delegates. Although the Agile series and its predecessors attract international participants, they have always been held in North America, so most of the delegates represent that continent.

Figure 2 shows the Agile 2008 poll results. We sampled 72 delegates, just over 4 percent of the total attendance. Although this number isn't quite above the magic 5 percent mark, it's still good enough for the results to be meaningful. The profile in terms of participants' roles is almost the inverse of ICSE, with 81 percent of the respondents identifying themselves as practitioners and 38 percent identifying as researchers. Again these groups overlapped, including the 22 percent of the respondents who identified themselves as both practitioners and

A Note Regarding the Cover

Because the IEEE Computer Society is dedicated to environmental responsibility, we've asked our printer to change from a UV-coated cover stock to an aqueous coating. The ultraviolet lamps used in the UV coating process require additional energy, and the chemicals in the UV coating not only raise worker safety issues at the printing plant but also prevent efficient recycling. An aqueous coating, on the other hand, can be freely recycled and re-pulped without harmful byproducts. This is a modest but effective step toward minimizing the environmental impact of our publications.

Mission Statement:

To be the best source of reliable, useful, peer-reviewed information for leading software practitioners—the developers and managers who want to keep up with rapid technology change.

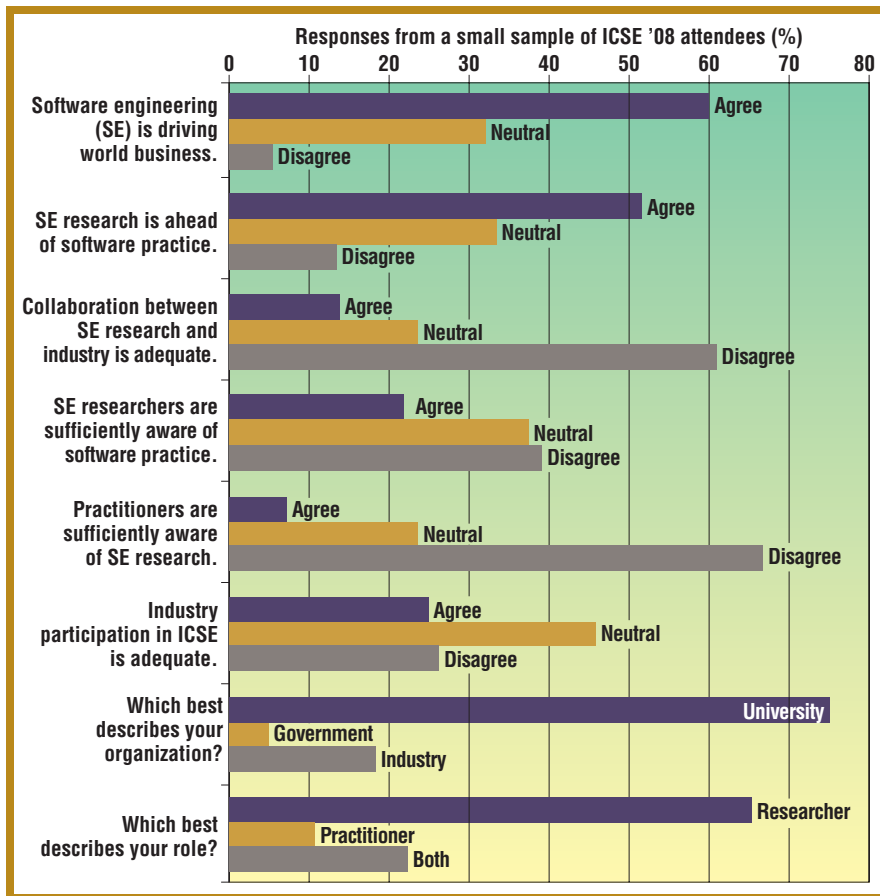


Figure 1. ICSE poll results. Respondents almost unanimously thought that practitioners are insufficiently aware of software engineering research.

researchers. Respondents' affiliations presented a somewhat more homogeneous picture than those of ICSE, with a distribution of 82 percent industry, 3 percent government, and only 11 percent academia.

In general, Agile attendees were more evenly divided in their opinions than ICSE attendees. For the first six questions (the ones on perceptions), clear majorities are altogether absent in the Agile version of the poll, whereas they're dominant in the ICSE version. So despite the fact that ICSE is more diverse in terms of the attendees' geographical origins, it appears to be more uniform than Agile in terms of cultural beliefs. Could this be because the research culture tends to be more uniform? If that's so, then perhaps culture trumps geographical diversity, and we shouldn't expect geographical diversity to skew the observations too much.

Starter Questions

ICSE's 2008 motto was "Driving World Business." The poll used this motto both as an ice-breaker and to probe the delegates'

confidence in their field's impact on industry. So, my volunteer asked the respondents whether they really thought software engineering was driving world business. The majority, 61 percent, happily agreed, and only 5 percent disagreed. However, a significant portion—one third of the respondents—weren't sure.

For the Agile version of the poll, I changed the first question to evaluate the delegates' interest in that conference's research tracks. Remarkably, over one-third indicated that they were planning to attend or had already attended at least one research track. The rest either didn't know or didn't plan to, in equal proportions. What's even more encouraging is that the percentages don't change much when projected onto respondents who identified themselves as practitioners. There was anecdotal evidence of this cross-interest as well. Indeed, two research sessions that I observed were well attended, and a large preconference workshop dedicated to research in progress was jam packed and very lively. Research tracks seemed

Hakan Erdogmus

hakan.erdogmus@computer.org

EDITOR IN CHIEF EMERITUS:

Warren Harrison, Portland State University

ASSOCIATE EDITORS IN CHIEF

Computing Now: Maurizio Morisio, Politecnico di Torino; maurizio.morisio@polito.it

Design/Architecture: Uwe Zdun, Vienna Univ. of Technology; zdun@infosys.tuwien.ac.at

Development Infrastructures: Martin Robillard, McGill University; martin@cs.mcgill.ca

Distributed and Enterprise Software:

John Grundy, University of Auckland; john-g@cs.auckland.ac.nz

Empirical Results: Forrest Shull, Fraunhofer Center for Experimental Software Engineering, Maryland; fshull@fc-md.umd.edu

Human and Social Aspects: Helen Sharp, The Open University, London; h.c.sharp@open.ac.uk

Management: John Favaro,

INTECS; john@favaro.net

Processes and Practices: Frank Maurer, University of Calgary; maurer@cpsc.ucalgary.ca

Programming Languages and Paradigms:

Laurence Tratt, Bournemouth University; laurie@tratt.net

Quality: Annie Combelles, DNV/Q-Labs; annie.combelles@dnv.com

Requirements: Ann Hickey, University of Colorado at Colorado Springs; ahickey@uccs.edu

DEPARTMENT EDITORS

On Architecture: Grady Booch, IBM

Bookshelf: Art Sedighi, SoftModule

Career Development: Philippe Kruchten, University of British Columbia

Design: Rebecca J. Wirfs-Brock, Wirfs-Brock Associates

Loyal Opposition: Robert Glass, Computing Trends

Requirements: Neil Maiden, City University London

Software Technology: Christof Ebert, Vector

Tools of the Trade: Diomidis Spinellis, Athens Univ. of Economics and Business

User Centric: Jeff Patton, consultant

Voice of Evidence: Forrest Shull, Fraunhofer Center for Experimental Software Engineering

ADVISORY BOARD

Frances Paulisch, Siemens (Chair)
Jennitta Andrea, ClearStream Consulting
Elisa Baniassad, Chinese University of Hong Kong

J. David Blaine, consultant

Kaoru Hayashi, SRA

Simon Helsen, IBM Rational

Juliana Herbert, Herbert Consulting

Gargi Keeni, Tata Consultancy Services

Karen Mackey, Cisco Systems

Steve McConnell, Construx Software

Erik Meijer, Microsoft

Grigori Melnik, Microsoft

Bret Michael, Naval Postgraduate School

Ann Miller, University of Missouri, Rolla

Deependra Moitra, consultant

Linda Rising, consultant

Wolfgang Strigel, consultant

Dave Thomas, Bedarra Research Labs

Markus Völter, consultant

STAFF

Senior Lead Editor
Dale C. Strok
dstrok@computer.org

Senior Editorial Services Manager
Crystal Shif

Magazine Editorial Manager
Steve Woods

Staff Editors
**Brian Brannon, Rebecca Deuel,
Jennifer Gardelle, Dennis Taylor, Linda World**

Assoc. Peer Review Manager
Hilda Carman

Publications Coordinator
Kathleen Henry
software@computer.org

Production Editor
Jennie Zhu

Technical Illustrator
Alex Torres

Director, Products & Services
Evan Butterfield

Digital Library Marketing Manager
Georgann Carter

Senior Business Development Manager
Sandra Brown

Senior Advertising Coordinator
Marian Anderson

CONTRIBUTING EDITORS

**Cheryl Baltes, Molly Gamborg,
Rita Scanlan, Joan Taylor**

CS PUBLICATIONS BOARD

Sorel Reisman (chair), Angela Burgess,
Chita R. Das, Van Eden, Frank E. Ferrante,
David A. Grier, Pamela Jones, Phillip A. Laplante,
Simon Liu, Paolo Montuschi, Jon Rokne,
Linda I. Shafer, Steven L. Tanimoto

MAGAZINE OPERATIONS COMMITTEE

David A. Grier (chair), David H. Albonesi,
Arnold (Jay) Bragg, Carl K. Chang,
Kwang-Ting (Tim) Cheng, Norman Chonacky,
Fred Douglas, Hakan Erdogmus, James A. Hendler,
Carl E. Landwehr, Dejan Mijolicic,
Sethuraman (Panch) Panchanathan,
Crystal R. Shif, Maureen Stone,
Roy Want, Jeffrey Yost

Editorial: All submissions are subject to editing for clarity, style, and space. Unless otherwise stated, bylined articles and departments, as well as product and service descriptions, reflect the author's or firm's opinion. Inclusion in IEEE Software does not necessarily constitute endorsement by the IEEE or the IEEE Computer Society.

To Submit: Access the IEEE Computer Society's Web-based system, Manuscript Central, at <http://cs-ieee.manuscriptcentral.com/index.html>. Be sure to select the right manuscript type when submitting. Articles must be original and not exceed 5,400 words including figures and tables, which count for 200 words each.

FROM THE EDITOR

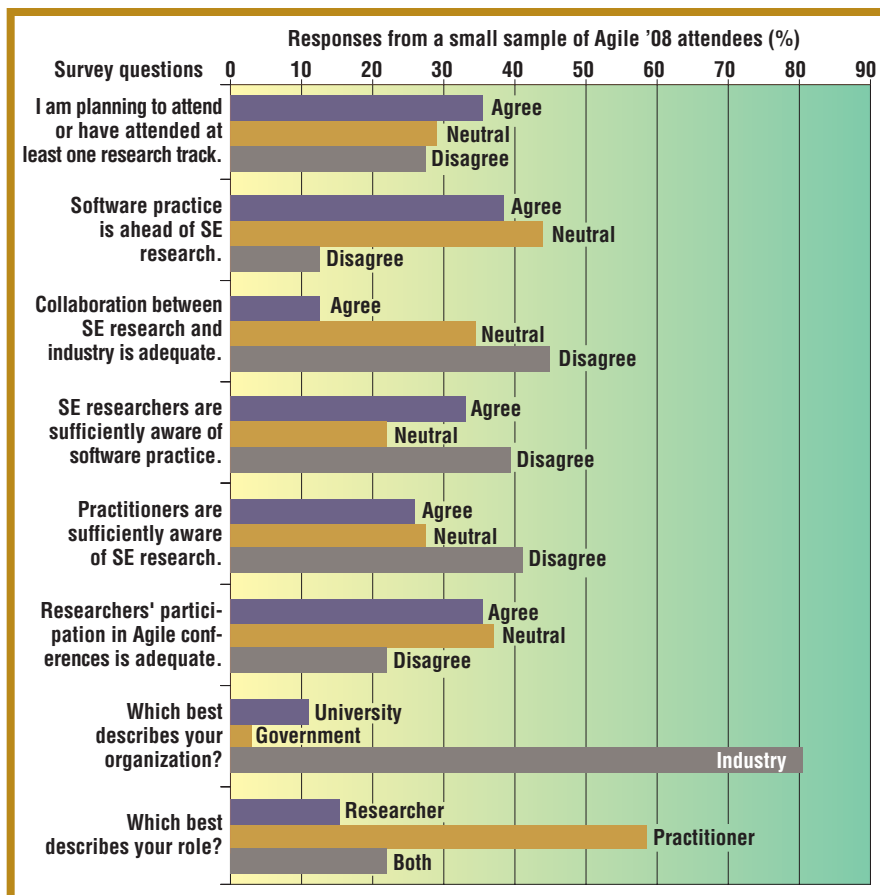


Figure 2. Agile poll results. Researchers who've been exposed to practitioners or who straddle both worlds have a higher opinion of software practice.

to bring variety and add value to the conference.

Who's Ahead, Who's Trailing?

The second question embodied my main motivation in conducting the polls: we asked respondents which community they thought was leading the pack. Fifty-three percent of ICSE respondents thought software engineering research is leading software practice, compared to 34 percent who thought otherwise. This result isn't too surprising to me given that ICSE is mainly a research conference. The percentages were similar for the group that contained researchers and the group that contained practitioners.

On the Agile side, 39 percent of the respondents thought software practice is leading software engineering research, and only 13 percent thought otherwise. Again, this is a rather unsurprising result. However, the responses aren't homogeneous between groups. The positive impression of software practice relative to software research increased from 39 percent to 52 per-

cent when we restricted responses to those who identified themselves as researchers. Researchers who've been exposed to the other side or who straddle both worlds overall have a higher opinion of software practice. This observation is particularly telling, especially because the converse doesn't appear to hold for ICSE attendees: practitioners at ICSE are in a similar situation to researchers at Agile, but the practitioner group doesn't have a higher opinion of software research as leading the pack than the average ICSE attendee.

The Need for More Collaboration

I'd like to think that neither community is ahead per se relative to the other. Researchers and practitioners sometimes tackle different problems. And when they do tackle similar problems, they often address them at different levels, with different senses of urgency, with different methods, and most important, from different perspectives. Such diversity is good for solving complex problems. Thus, I prefer to believe that

they have much to learn from each other, and this brings us to the inevitable subject of cross-collaboration. Not enough is happening on that front. And my own perceptions are echoed in the polls' results.

Respondents from both conferences who think that collaboration between software engineering research and industry is adequate are indeed a rather small minority, at 14 percent for ICSE and 13 percent for Agile. It's encouraging that both audiences acknowledge the need for more collaboration, but are they putting their feet where their mouths are? The acknowledgment is nice, but more widespread, real action would be even nicer.

Insufficient Mutual Awareness

In the same vein, admitting failure was also forthcoming as far as mutual awareness is concerned. The direction of the sentiment was homogeneous between the two conferences.

ICSE respondents who thought that the awareness level of software practice among software engineering researchers is inadequate outnumbered those who think it's adequate by nearly two to one. Interestingly, this proportion was lower—four to three—for Agile respondents, who apparently assumed a more even stance.

Conversely, ICSE respondents almost unanimously thought that practitioners are insufficiently aware of software engineering research. This group outnumbered those who thought otherwise by almost 10 to 1. Although the sentiment's direction was the same for Agile respondents, the proportions differed considerably. Agile respondents who thought that practitioners are insufficiently aware of software engineering research still outnumbered those who thought the opposite, but only by a ratio of less than two to one—again, a much less dramatic stance compared to ICSE.

The Importance of Cross-Participation

When asked about the level of industry participation in the conference, ICSE respondents took an overall neutral position, with only 25 percent believing the level was adequate, slightly more believing it was inadequate, and the rest remaining indifferent.

The response was similar when we asked the Agile attendees the converse ques-

tion: 35 percent thought researchers' participation at Agile was adequate, 22 percent thought it was inadequate, and the rest were indifferent. This doesn't sound like a crowd cheering for more academic involvement in the conference.

So let's recap what people were saying: We need more collaboration. We need more mutual awareness. But we don't care about having more of the other side at our conference. Oh well.

Last Words

In the sampled audiences overall, practitioners' impressions of software engineering research were less negatively biased than researchers' impressions of software development practice. More cross-pollination was deemed desirable in general, but whether anything is to be done about it wasn't clear to either audience.

Nonetheless, inaction doesn't suit the organizers who will shape the two conferences in the near future. Surely, both conferences will continue to deliver their usual high-quality content for their respective main audiences. On top of that, ICSE

'09 is gearing up for a stronger-than-ever industry track, led by *IEEE Software's* very own Frances Paulisch and Philippe Kruchten. Reciprocally, expect similar efforts from Agile '09 for a beefed-up research program led by organizing committee members Tore Dyba, Yael Dubinsky, and Philippe Kruchten.

If you know how other software engineering and developer conferences encourage interaction between researchers and practitioners, please write to me at hakan.erdogmus@comptuer.org. 

Acknowledgments

I am grateful to volunteers Jennifer Schiller and Chandra Sreeraman for conducting the polls and to Frances Paulisch and Philippe Kruchten for their comments on an earlier version of this essay. Also, thanks to Frances Paulisch, Grigori Melnik, Rachel Davies, and Ahmed Sidky, who were members of the two conferences' organizing committees, for giving me permission to do the polls, providing logistic support, and securing Jennifer and Chandra's help.

Classified Advertising

SUBMISSION DETAILS: Rates are \$110.00 per column inch (\$125 minimum). Eight lines per column inch and average five typeset words per line. Send copy at least one month prior to publication date to: Marian Anderson, Classified Advertising, *IEEE Software Magazine*, 10662 Los Vaqueros Circle, PO Box 3014, Los Alamitos, CA 90720-1314; (714) 821-8380; fax (714) 821-4010. Email: manderson@computer.org.

ROSE-HULMAN INSTITUTE OF TECHNOLOGY. The Department of Computer Science and Software Engineering at Rose-Hulman Institute of Technology invites applications for a visiting faculty position (pending budget approval) for the 2009-2010 academic year. We are especially interested in candidates who can contribute to our software engineering expertise in one or more of the following areas: software project management, software architecture, software testing and quality assurance, software maintenance and evolution, and software construction. An advanced degree in

computer science, software engineering, or a closely related field is normally required of applicants from academia; exceptions will be made for candidates with appropriate industry experience. Applicants with significant software engineering experience are particularly encouraged to apply. Rose-Hulman is a highly selective, coeducational, primarily undergraduate college of engineering and science, with a vision to be the best institution of its kind in the world. Faculty members are expected to be outstanding teachers and professionally active. The department offers B.S. degrees in software engineering and computer science. These programs are accredited by the Engineering Accreditation Commission and the Computing Accreditation Commission, respectively, of ABET, Inc. Program descriptions may be found in the "Academics" area of www.rose-hulman.edu. Applicants should submit a cover letter, curriculum vitae, and a statement on their teaching philosophy to: <https://jobs.rose-hulman.edu>. Three letters of recommendation should be submitted by the applicant's references through the same website. EEO/AA.